Name: _____

WORKSHEET

Equations of circles

1	17	13	18	14	17	13		5	3	18	14	17		15	8	2	4	15	12	2										
14	18	13	7	18	5	15	18	11	17					10	17	4	3	12				4	11	14						
9	18	11	14					7	8	11	6	18	14	17	11	7	17					18	13							
12	10	17				16	17	2				12	8				4	15	15			12	10	17						
15	8	7	16	13																										

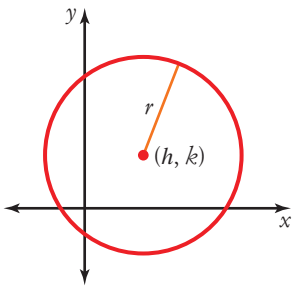
Joe Paterno

Rule

The equation of a circle is

$(x - h)^2 + (y - k)^2 = r^2$

where (h, k) is the centre of the circle and r is the radius.



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Questions

Find the centre and radius of each of the following circles.

1 $(x + 3)^2 + (y - 1)^2 = 9$

2 $(x - 2)^2 + (y + 10)^2 = 16$

3 $(x + 6)^2 + (y - 5)^2 = 4$

4 $(x - 5)^2 + (y - 2)^2 = 49$

5 $(x + 1)^2 + (y - 8)^2 = 36$

Give the equations for the circles with the following centres and radii.

6 Centre $(2, -7)$, radius $= 4$

7 Centre $(-1, 4)$, radius $= 1$

8 Centre $(-5, -6)$, radius $= 7$

9 Centre $(8, 3)$, radius $= 3$

10 Centre $(11, -2)$, radius $= 5$

Find the centres and radii of the following circles.
(Hint: Use 'complete the square' to factorise first).

11 $x^2 + y^2 - 6x + 2y + 1 = 0$

12 $x^2 + y^2 + 4x - 14y + 28 = 0$

13 $x^2 + y^2 - 10x - 4y + 13 = 0$

14 $x^2 + y^2 + 8x - 6y + 21 = 0$

15 $x^2 + y^2 - 2x - 8y - 8 = 0$

Sketch the following circles.

16 $(x - 6)^2 + (y + 2)^2 = 25$

17 $x^2 + y^2 - 8x - 6y + 24 = 0$

18 $x^2 + y^2 + 2x + 4y - 4 = 0$

Solutions

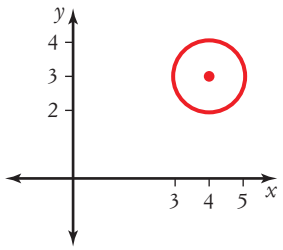
A Centre $(5, 2)$, radius $= 7$

B Centre $(-3, 1)$, radius $= 3$

C $(x + 1)^2 + (y - 4)^2 = 1$

D Centre $(-4, 3)$, radius 2

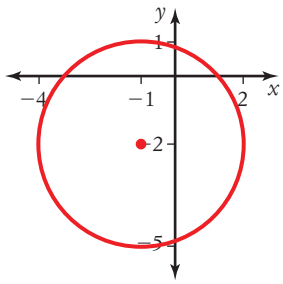
E



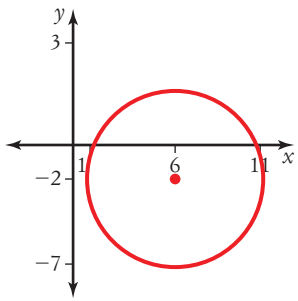
F $(x - 2)^2 + (y + 7)^2 = 16$

H $(x - 11)^2 + (y + 2)^2 = 25$

I



K



L Centre $(1, 4)$, radius $= 5$

M $(x - 8)^2 + (y - 3)^2 = 9$

N Centre $(3, -1)$, radius $= 3$

O $(x + 5)^2 + (y + 6)^2 = 49$

P Centre $(-1, 8)$, radius $= 6$

R Centre $(-6, 5)$, radius $= 2$

S Centre $(5, 2)$, radius $= 4$

T Centre $(-2, 7)$, radius $= 5$

Y Centre $(2, -10)$, radius $= 4$

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